Data Visualization and Business Intelligence with Tableau Desktop

1

Data Visualization

Core Concepts and Principles for Technical Users

2026

2

Why Data Visualization Matters

Improvement in understanding and recall with proper color combinations

Men with color vision deficiency

- **Faster decision-making**: Visual patterns emerge instantly from datasets, allowing teams to spot trends in seconds rather than hours
- **Clear communication**: Complex information becomes accessible visually, ensuring stakeholders interpret data consistently and reducing miscommunication
- **Team alignment**: Shared visual dashboards create common understanding of metrics, enabling teams to coordinate data-driven decisions
- **Quality assurance**: Visual representations reveal data anomalies and outliers hidden in spreadsheets, helping maintain data integrity throughout pipelines

3

Understanding Your Audience

Tailoring visualizations to audience needs determines whether insights drive action or get ignored

- **Executives need impact**
  Focus on bottom-line insights with clean bar charts, trend lines, and big numbers showing ROI, key trends, and actionable recommendations—avoid technical jargon and detailed methodologies [1]
- **Analysts need depth**
  Provide methodology transparency with scatter plots, box plots, detailed multi-series charts, confidence intervals, and data tables to support rigorous statistical validation [1]
- **General audiences need clarity**
  Use simple comparisons, familiar references, progressive disclosure, and clear annotations with infographics and interactive elements to make insights relatable and engaging [1]
- **Technical teams need specifications**
  Show workflows, highlight bottlenecks, include benchmarks with process flows, performance dashboards, and diagnostic charts focused on operational metrics and implementation details [1]

4

Comparison
Bar charts for category comparisons across same units

↓

Trends
Line charts for continuous time series with 4+ points

↓

Composition
Pie charts for parts-of-whole with 3-6 slices

↓

Correlation
Scatter plots for relationships between two variables

↓

Distribution
Histograms for frequency across value ranges

Choosing the Right Chart Type

- **Match chart to data question**: Comparison needs bar charts, trends need line charts, relationships need scatter plots, composition needs pie charts, and distribution needs histograms
- **Common mistakes to avoid**: Don't use pie charts for more than 6 slices, don't use bar charts for time series, don't use line charts for non-continuous categorical data
- **Decision framework**: Start by identifying whether you're comparing values, showing composition, understanding distribution, or analyzing trends to select the optimal visualization type

5

Chart Types by Use Case

Chart Type	Best Use Case
Bar	Comparing discrete categories, 3-8 items, same units
Line	Trends over time, 4+ evenly-spaced points
Area	Cumulative trends with volume emphasis
Pie/Doughnut	Parts of whole, 3-6 slices totaling 100%
Scatter	Correlation between variables, 10+ points
Bubble	Three-dimensional comparison with X-Y-size
Radar	Multi-attribute comparison, 5-8 attributes
Table	Mixed units or detailed specifications

Source: Different Chart Types Explained: When to Use Each (wi...

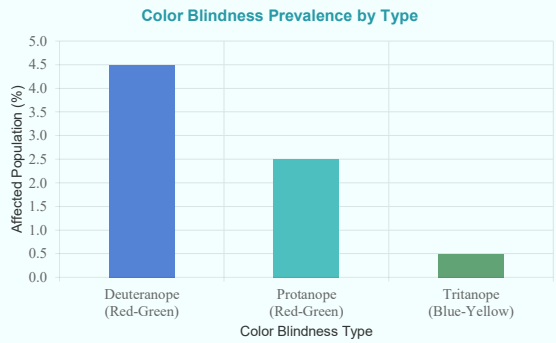
Chart Type Reference Guide

- **Bar vs column distinction:** Horizontal bars work best for long category labels and comparing more than 10 items, while vertical columns excel at showing hierarchical or time-based categorical data
- **When to use tables:** Choose tables over charts when data has different units, scale ratios exceed 10:1, or you need to show detailed specifications with multiple attributes per item
- **Specialized chart applications:** Waterfall charts display cumulative effects, Gantt charts show project timelines, heatmaps reveal density patterns, and box plots illustrate distribution statistics

6

Color Theory Fundamentals

- **Color accessibility imperative**: Approximately 8% of men and 0.5% of women have color vision deficiency, making accessible color choices essential for inclusive design
- **Avoid problematic combinations**: Never use red-green, green-brown, green-blue, or blue-gray pairings as they are indistinguishable to people with common color blindness types
- **Use colorblind-friendly palettes**: Prefer blue-orange, yellow-purple combinations and test visualizations with simulators before publishing [8]
- **Add redundancy beyond color**: Incorporate textures, patterns, symbols, shapes, or labels to reinforce groupings for accessibility



Source: Accessible Data Visualizations | Principles of Data...

7

Managing Cognitive Load

Reducing cognitive load through design enables viewers to focus mental resources on understanding insights rather than decoding visuals

5-7 items
Human working memory capacity limit

3 types
Intrinsic, germane, extraneous cognitive load

- **Intrinsic load is inherent complexity**: The dataset's natural complexity cannot be eliminated—learning calculus requires more cognitive effort than basic addition regardless of presentation method
- **Germane load reflects audience familiarity**: The same visualization appears different to novices versus experts based on their existing mental frameworks and domain knowledge
- **Extraneous load from poor design**: Clutter, inconsistent formatting, and unnecessary visual elements waste mental resources that should be focused on interpreting data insights
- **Design to minimize extraneous load**: Limit dashboards to 5-7 key metrics, remove non-essential decorations, maintain consistent color schemes, and ensure clear visual hierarchy to optimize comprehension

8

Dashboard Design Principles

Effective dashboards answer specific questions with 5-7 focused metrics, context, and actionable insights

5-7 KPIs
Optimal dashboard metric count

80%
Dashboards unused due to overwhelming users [10]

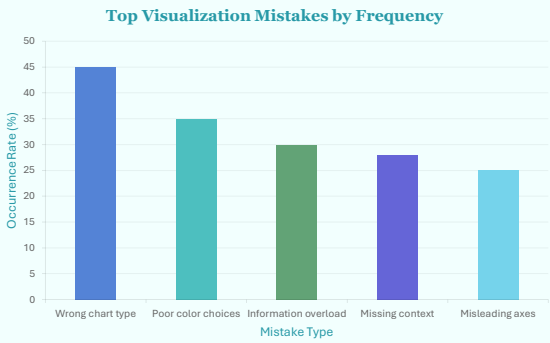
- **Start with a specific question:** Every dashboard should answer a clear decision-focused question like "Are we on track for quarterly targets?" rather than attempting to show all available data
- **Limit to 5-7 KPIs maximum:** Human working memory holds 5-7 items—dashboards with 20 metrics get ignored while those with 6 are actionable
- **Context matters more than numbers:** A metric without comparison is meaningless—always include targets, benchmarks, trends, or prior periods
- **Use predictable layout patterns:** Position critical metrics top-left following natural eye scanning, group related data together
- **Design for real-time actionability:** Include status indicators, clear ownership, and consistent refresh cadence so every metric drives decisions

9

Best Practices Checklist

- **Chart selection verification**: Match chart type to data structure—bar for comparisons, line for trends, scatter for correlations, pie for composition, table for mixed units
- **Color accessibility check**: Test all visualizations with colorblindness simulators, avoid red-green combinations, ensure 4.5:1 contrast ratio minimum, and add non-color redundancy
- **Cognitive load assessment**: Limit metrics to 5-7 per view, remove unnecessary visual elements, maintain consistent formatting, and ensure clear visual hierarchy
- **Layout optimization**: Apply predictable patterns like F-shaped or Z-shaped layouts, position critical information top-left, and group related metrics together
- **Context provision**: Include targets, benchmarks, time periods, and comparisons so viewers understand whether performance is good or bad
- **Simplicity enforcement**: Start Y-axis at zero for bar charts, use clear labels, limit to 2 font families, and avoid 3D effects or unnecessary decoration
- **Quality validation**: Verify data accuracy, ensure high-resolution images, test on different devices, and confirm all metrics have clear ownership and update frequency

10



Sources:
10 Essential Data Visualization Best Practices for 2025
Data Visualization Tips and Best Practices - Tableau
Data Visualization Colors: Best Practices & Palettes ...
Cognitive Load as a Guide: 12 Spectrums to Improve Yo...
Effective Dashboard Design: Principles, Best Practice...

Common Pitfalls to Avoid

Recognizing and avoiding common visualization mistakes prevents miscommunication and ensures data integrity

- **Choosing wrong chart types**: Using pie charts for trends, bar charts for correlations, or line charts for discrete categories obscures insights
- **Overloading with information**: Cramming 20+ metrics onto one dashboard or using too many colors reduces comprehension and prevents action
- **Ignoring accessibility**: Failing to test for color blindness or relying solely on color excludes significant portions of your audience
- **Manipulating scale or axes**: Truncated Y-axes or inconsistent intervals mislead viewers and damage credibility of insights

11

References

[1] Principles of Effective Data Visualization
[2] 10 Essential Data Visualization Best Practices for 2025
[3] Data Visualization Tips and Best Practices - Tableau
[4] Different Chart Types Explained: When to Use Each (wi...
[5] Choosing the Right Chart Type: A Technical Guide
[6] Data Visualization Colors: Best Practices & Palettes ...
[7] The Role Of Color Theory In Data Analysis And Visuali...
[8] 6 Accessible Data Visualizations | Principles of Data...
[9] Cognitive Load as a Guide: 12 Spectrums to Improve Yo...
[10] Effective Dashboard Design: Principles, Best Practice...

12

Business Intelligence

A Technical Overview for IT Professionals and Analysts

13

What Is Business Intelligence?

79%
of enterprises operate with fragmented data systems

$13B
BI market projected to reach by 2025

- BI is a set of processes for collecting and analyzing data to yield insights that inform business strategies
- It enables access to historical and current data from multiple sources including third-party systems and databases
- BI provides a foundation for reporting and analytics by transforming raw data into meaningful insights
- The core purpose is helping organizations make informed choices based on actual data rather than assumptions

14

BI vs Data Analytics - Understanding the Distinction

Business Intelligence:

- Focuses on descriptive and diagnostic insights to understand what happened and why using historical structured data
- Delivers dashboards, reports, and visualizations that support strategic planning and operational decision-making across the organization
- Provides a single version of truth by aggregating data from CRM, ERP, and marketing platforms into unified visual insights
- Empowers business users with self-service tools for quick exploration and reporting without requiring advanced technical skills

Data Analytics:

- Uses statistical and computational methods to predict outcomes and prescribe future actions based on data trends and patterns
- Explores both structured and unstructured data to uncover patterns, correlations, and predictive insights using advanced analytical methods
- Goes beyond reporting to answer "why" questions through root cause analysis and "what if" scenario modeling
- Requires specialized analytical skills and tools like Python, R, and machine learning platforms for deep investigation

15

มิติ	Business Intelligence (BI)	Data Analytics
โฟกัสหลัก	อดีต & ปัจจุบัน	อนาคต & การคาดการณ์
ประเภทคำตอบ	What / Why (เชิงอธิบาย)	Why / What if / What next
ข้อมูล	Structured (ERP, CRM)	Structured + Unstructured
ผลลัพธ์	Dashboard, Report	Model, Prediction, Scenario
ผู้ใช้งานหลัก	ผู้บริหาร, ผู้ใช้ธุรกิจ	Data Analyst, Data Scientist
ความซับซ้อน	ปานกลาง	สูง

16

Core Components of BI Architecture

- Data sources include software, systems, or devices producing structured, semi-structured, or unstructured data from CRMs and external sources [5]
- Data integration layer uses ETL pipelines to ensure data completeness, accuracy, and consistency by extracting, transforming, and loading [5]
- Data storage includes warehouses for historical data, marts for department-specific needs, and operational stores for real-time access
- BI and analytics layer provides tools for querying, analysis, and delivering insights through reports, dashboards, and visualizations

Data Sources
Internal and external systems generating business data

Data Integration
ETL processes extracting, transforming, and loading data

Data Storage
Warehouses and marts organizing information for analysis

Analytics Layer
Tools for querying, analyzing, and processing data

Visualization
Dashboards and reports presenting insights to users

Governance
Security, quality, and compliance management controls

17

Extract
Pull data from multiple disparate sources and systems

↓

Transform
Clean, standardize, and enrich data for consistency

↓

Load
Insert processed data into warehouse for analysis

The ETL Process - Extract, Transform, Load

54%
faster time-to-insight

67%
reduction in errors vs manual

- Extraction collects raw data from CRMs, ERPs, cloud services, APIs, and IoT platforms
- Transformation cleans, deduplicates, and standardizes data for consistency across sources
- Loading moves data into central warehouse for reporting and analytics access
- ETL ensures data quality and provides reliable foundation for BI activities

18

Characteristic	Data Warehouse	Operational Database
Purpose	Analytics & Reporting	Daily Transactions
Data Type	Historical & Current	Current Only
Optimization	Complex Queries	Fast Reads/Writes
Structure	Denormalized Schema	Normalized Tables
Update Frequency	Batch (Periodic)	Real-time (Continuous)

Data Warehousing Fundamentals

- A data warehouse is a central platform for consolidating and storing data from different sources, preparing it for downstream BI and analytics [4]
- Data warehouses store historical and current data in structured formats optimized for complex querying and connect to BI tools for reports and visualizations
- Data marts are subsets of data warehouses tailored to specific departments, allowing teams to quickly access relevant information for analytics
- Unlike operational databases designed for transactions, warehouses use denormalized schemas for fast query performance on large datasets

19

Key BI Concepts - Data Assets and Repositories

Modern BI leverages multiple types of data repositories, each optimized for specific use cases and analytical needs

Data Lake: Raw storage for all data types

Data Mart: Department-specific subset

ODS: Real-time operational reporting

- **Data Warehouse:**
  Organized repository where data from various sources is versioned and structured for easier reporting and analysis, serving as the authoritative source [4]
- **Data Mart:**
  A focused section of the data warehouse containing information for specific areas like sales or finance, enabling departments to quickly access the information they need
- **Data Lake:**
  Large storage system holding wide range of data types in raw original form, supporting both structured tables and unstructured data like text or images [4]
- **ODS (Operational Data Store):**
  System for operational reporting managing day-to-day business activities with current data only, perfect for real-time reporting without historical records [4]
- **Data Vault:**
  Design approach for data warehouses creating flexible and scalable systems that handle changes and updates over time efficiently

20

The Role of BI in Decision-Making

23x more likely to acquire customers with data-driven workflows

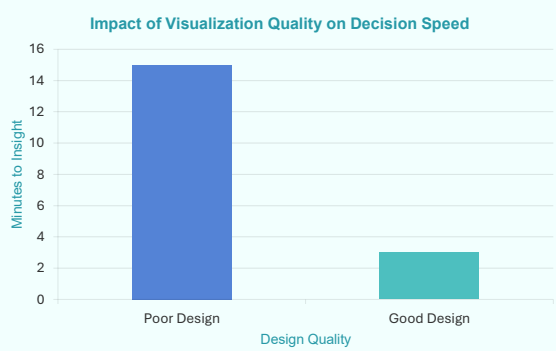
6x more likely to retain customers

- BI provides tools that help decision-makers work smarter by answering key questions with data insights
- Organizations monitor performance against goals through real-time dashboards and KPIs that track critical metrics
- BI identifies trends in large datasets enabling deeper understanding of performance and operational efficiency
- Predictive analytics guide product development, inventory optimization, and strategic planning decisions

21

Data Visualization and Reporting Best Practices

- Choose the right chart type by matching visualization format to the analytical goal
- Maintain high data-ink ratio by eliminating unnecessary visual elements and keeping designs simple
- Design dashboards with logical layout following the three-second, thirty-second, three-hundred-second rule
- Ensure consistent metric definitions across all dashboards so every visualization tells the same truth



Source: Data Visualization Best Practices for Better Decision...

22

Modern BI Platforms and Self-Service Analytics

Traditional BI Characteristics:

- IT-driven approach with analysts generating structured reports from internal databases
- Slow data analysis as users wait for IT processing, creating decision bottlenecks
- Best suited for large enterprises with well-defined structures and infrequent changes

Modern BI Capabilities:

65%

of enterprises regularly use AI in at least one business function

- Self-service analytics with drag-and-drop interfaces for business users
- AI-driven features including natural language queries and automated insights
- Cloud-based platforms with real-time access and scalable infrastructure
- Interactive dashboards enabling drill-down and dynamic filtering

23

Define Objectives
Identify key business questions and decision needs

↓

Design Architecture
Plan data sources, integration, storage, and delivery

↓

Select Tools
Choose BI platforms matching technical requirements

↓

Implement Governance
Establish data quality and security controls

↓

Iterate
Continuously refine based on user feedback

Getting Started with BI Implementation

Successful BI implementation requires organizational buy-in, clear objectives, proper architecture design, and continuous iteration aligned with business goals

- Start by understanding what business questions need answering and design reports that deliver those answers without ambiguity
- Select appropriate architecture type - centralized for control, decentralized for autonomy, hybrid for balance, or cloud-based for scalability [3]
- Choose BI tools based on number of connectors, cloud compatibility, ease of use for business users, and total cost of ownership [3]
- Implement strong data governance from the start including consistent naming conventions, clear metric definitions, and sound data quality controls

24

References

[1] What is business intelligence (BI)? - IBM

[2] 8.2 Foundations of Business Intelligence and Analytic...

[3] What is BI Architecture? The Complete Guide - Thought...

[4] Business Intelligence Concepts, Components and Applic...

[5] Business Intelligence Architecture: Components & Top ...

[6] Business Intelligence Vs Data Analytics: 15 Key Diffe...

[7] Data Visualization Best Practices for Better Decision...

25

26

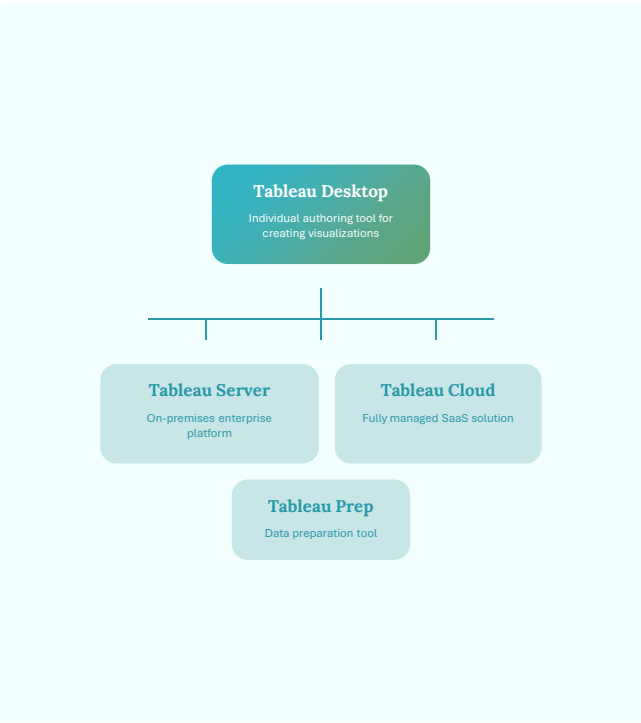
Why Tableau Desktop?

Tableau Desktop transforms raw data into actionable insights through powerful visual analytics that anyone can use [6]

24-month	350+
support lifecycle	native data connectors

- Comprehensive data visualization application enabling interactive graphs, dashboards, and reports within minutes
- Intuitive drag-and-drop interface accessible to both technical and non-technical users
- Connects to diverse data sources from spreadsheets to enterprise warehouses
- Cross-platform support for Windows and macOS with Unicode compatibility

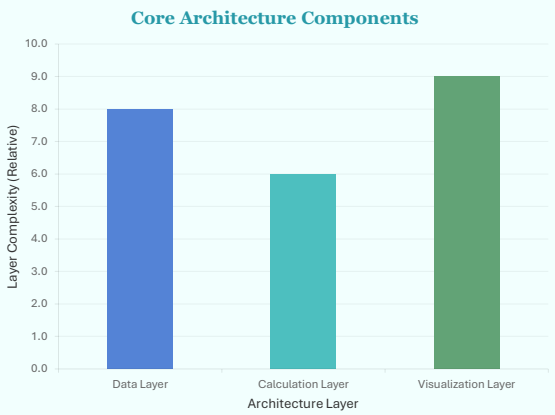
27



The Tableau Product Ecosystem

- Tableau Desktop functions as the primary development environment where analysts connect to data and create visual dashboards
- What you create in Desktop can be published to Server or Cloud, enabling secure distribution across the organization
- The integration between Desktop and Server creates a complete analytics workflow from individual exploration to enterprise collaboration

28



Source: A Detailed Guide to Tableau Architecture: Desktop and...

Desktop Architecture Overview

- Data Layer manages connections to diverse sources through native connectors, supporting live connections and extracts
- Calculation Layer handles transformations, custom calculations, and aggregations without modifying source data
- Visualization Layer creates interactive dashboards using VizQL technology that translates actions into queries
- Layered architecture separates concerns while maintaining flexibility for diverse analytical requirements

29

Comprehensive Data Connectivity

Excel, CSV, JSON, PDF files

SQL Server, Oracle, MySQL databases

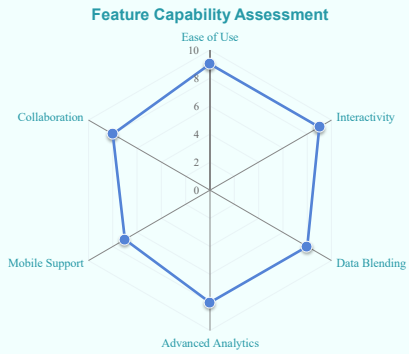
Snowflake, BigQuery, Redshift warehouses

- Native connectors built for databases, cloud warehouses, and file types from Excel to PDFs
- Live connections query sources in real-time, ideal for up-to-date dashboards
- Extract mode imports data into Hyper engine for faster querying and offline access
- ODBC, JDBC, and Connector SDK enable integration with any data source

30

Key Features and Capabilities

- Drag-and-drop interface enables rapid visualization creation without coding, allowing analysts to focus on insights
- Interactive dashboards update in real-time with filters, parameters, and drill-down capabilities empowering user exploration
- Data blending seamlessly combines multiple sources, uncovering correlations that single-source analysis would miss
- Advanced analytics integration with R, Python, MATLAB extends capabilities to machine learning and predictive analytics



Sources:
Tableau Features to Know from A to Z
Important Key Features of Tableau - Scaler Topics

31

Building Your First Visualization

- Connection starts on Start page, selecting file types, databases, or servers, then providing credentials
- Exploration uses drag-and-drop workspace where dimensions create structure and measures provide quantitative analysis
- Refinement applies filters to focus data, creates calculated fields, and adjusts formatting for clarity
- Dashboard assembly combines worksheets with interactive actions and filters to tell complete data stories

Connect
Link to data sources

↓

Explore
Drag dimensions and measures

↓

Refine
Apply filters and calculations

↓

Dashboard
Combine multiple views

↓

Share
Publish to Server or Cloud

32

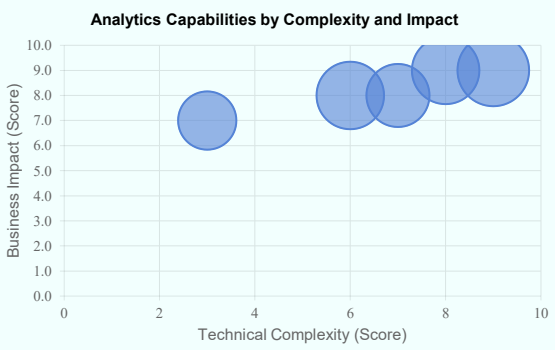
Best Practices for IT Implementation

24-month
support lifecycle

Enterprise-grade
security

- Implement data governance policies defining access controls, naming conventions, and templates for consistent visualizations
- Use extracts strategically for large datasets to improve performance and reduce production load
- Plan authentication integration with Active Directory, SAML, or OAuth for secure auditable access
- Deploy consistent versions, test in development, and maintain documentation for custom connectors

33



Sources:

Tableau Features to Know from A to Z
Tableau Desktop | Connect, analyze, and visualize any...

Advanced Analytics Capabilities

- Built-in statistical functions include trend lines, forecasting, and clustering without coding
- Analytics Extensions integrate R, Python, MATLAB, and Einstein Discovery for custom models
- Table calculations enable running totals and year-over-year comparisons without modifying data
- Parameters create interactive scenarios for immediate impact visualization and sensitivity analysis

34

Getting Started Resources

Official Training

Free online tutorials, certification programs, and instructor-led courses from beginner to advanced analytics

Community Forum

Connect with thousands of users worldwide sharing tips, solutions, and best practices for implementation

Getting Started Tutorial

Walk through connecting data, building visuals, creating dashboards using realistic business scenarios

Free 14-day trial available

Tableau Public

Free access with millions of published examples demonstrating best practices across industries

100+ accelerator templates

35

References

[1] Tutorial: Get Started with Tableau Desktop

[2] 26 Tableau Features to Know from A to Z

[3] A Detailed Guide to Tableau Architecture: Desktop and...

[4] All Technical Specifications - Tableau

[5] Connect to Your Data - Tableau

[6] Tableau Desktop | Connect, analyze, and visualize any...

[7] 10 Important Key Features of Tableau - Scaler Topics

36

Data Visualization with Tableau Desktop

Master the Essential Skills for Connecting, Organizing, and Analyzing Data

2026

37

Why Tableau Desktop Matters

What Tableau Does

Wide Data Support

Interactive Dashboards

- Connects to diverse data sources including Excel, CSV, databases, and cloud platforms without modifying original data
- Creates interactive visualizations that reveal patterns, trends, and outliers in your data through drag-and-drop interface
- Enables data exploration at multiple levels of detail from high-level summaries to granular record-level analysis

Your Learning Path

- You'll learn to connect to various data sources and establish live or extracted connections for analysis
- Master filtering and sorting techniques to focus on relevant data and identify key insights quickly
- Understand date hierarchies and time-based analysis to track trends and patterns over different periods

38

Understanding Tableau's Interface

The Tableau workspace is organized into logical areas that make data analysis intuitive and efficient

- The start page provides quick access to data connections with file types listed first, then server types and saved sources
- Dimensions appear in blue and typically contain categorical data like names, dates, and geographic information
- Measures appear in green and contain numerical data that Tableau automatically aggregates using functions like sum or average
- The Show Me panel suggests appropriate visualization types based on the fields you've selected for analysis

Connect Pane

Lists available data connectors for files, servers, and saved data sources

Data Window

Displays all fields from connected sources organized into dimensions and measures

Shelves Area

Rows, Columns, Filters, and Marks shelves where you drag fields to build views

Canvas

Main workspace where visualization appears and updates as you modify fields

39

Connecting to Your First Data Source

- Select Data > New Data Source or press Ctrl+D to open the connection dialog and choose your data type
- Tableau supports Excel, CSV, JSON, PDF, databases like SQL Server and MySQL, and cloud platforms including Google BigQuery
- After connecting, data fields display in the Data window where you can review field names, data types, and sample values
- The connection is live by default, meaning changes to source data automatically reflect in Tableau, or create an extract for faster performance

Steps to Connect Data in Tableau

1. Select data source type (Excel, CSV, Database) from Connect pane
2. Browse and select your data file or enter connection details
3. Preview data in Data Source tab and verify field types
4. Click on worksheet to begin building visualizations

Source: Connect to a sample data source - Tableau

40

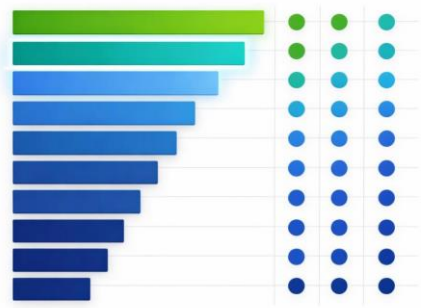
Sorting Data for Clear Insights

Sorting arranges data in meaningful order to reveal relationships, trends, and top performers

3 Click Types

Multiple Sort Methods

- Hover over an axis or field label to reveal the sort icon, then click to sort ascending, descending, or clear
- One click sorts numerically descending, two clicks ascending, three clicks returns to default order
- For alphabetic sorting, one click gives ascending A-Z, two clicks descending Z-A, three clears
- Use the sort dialog for advanced options like sorting by different fields or custom orders



41

Organizing Data with Hierarchies

- Tableau automatically creates date hierarchies with levels from year down to second based on your data's granularity
- Click the plus sign on a dimension pill to expand the hierarchy and add the next level of detail to your view
- You can create custom hierarchies by dragging one dimension onto another in the Data pane to group related fields
- Hierarchies allow viewers to start with a high-level overview and progressively drill down into more detailed segments as needed

Geographic Hierarchy

Country → State/Province → City

Product Hierarchy

Category → Sub-Category → Product Name

Date Hierarchy

Year → Quarter → Month → Day

42

Relative Dates

Last 7 days, This month, Year to date filters update automatically [7]

Range of Dates

Select specific start and end dates for custom analysis periods [7]

Date Parts

Filter by months, quarters, or days across all years [7]

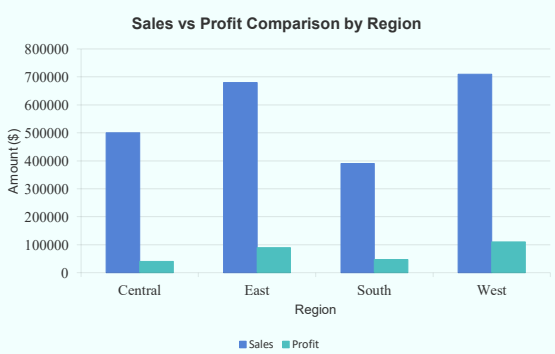
Special

Include today, null values, or non-null for data quality checks [7]

Slicing Data by Date Ranges

- Drag a date field to the Filters shelf to open the dialog with tabs for different filtering approaches
- Relative date filters like Last 30 Days or This Quarter automatically adjust as time passes, keeping dashboards current
- Range of Dates allows exact start and end dates for analyzing specific periods like fiscal quarters
- Combine date filters with Quick Filter controls to give viewers interactive control over time periods displayed

43



Source: Introduction to Tableau Desktop

Using Multiple Measures

Measure Names and Measure Values: Special auto-generated fields [4]

3 methods: Individual axes, Blended axis, Dual axis [4]

- Measure Values acts as a container holding multiple measures, while Measure Names provides the labels for distinguishing each metric [5]
- Drag measures to Rows or Columns shelf to create individual axes, allowing each measure to display with its own scale and visual encoding [4]
- Blend measures onto a single axis by dragging one measure onto another's axis, ideal when measures share similar scales and units [4]
- Dual axes layer two independent axes in the same pane, enabling comparison of measures with drastically different scales or units [4]

44

Building Multiple Measure Views

Measure Values field: Contains all measures combined [5]

Measure Names: Collection of all measure labels [5]

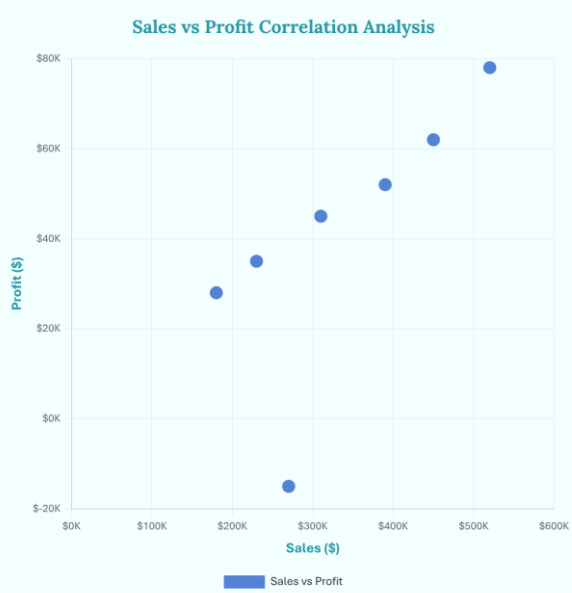
Side-by-side bars: Use Show Me panel option [5]

- For individual axes, place multiple measures on Rows shelf creating separate rows, each with independent scales perfect for measures with different units [4]
- Blended axis shows all measures in single pane on shared scale, automatically creating Measure Values field and filtering to selected measures only [4]
- When creating dual axes, right-click second measure and select Dual Axis, then synchronize axes if comparing similar ranges [4]
- Control which measures appear by dragging them in or out of the Measure Values card, and assign different colors using Measure Names on Color [5]

45

Scatter Plots for Relationships

- Place one measure on Columns shelf and another on Rows to create Cartesian coordinates where each dot represents data point intersection [6]
- Positive correlation appears when both variables increase together, negative when one increases as other decreases, and no correlation shows random scatter [7]
- Add dimensions like Category to Color or Shape on Marks card to segment data and identify which groups drive specific patterns [6]
- Trend lines provide statistical definition of relationships, with options for linear, logarithmic, exponential, or polynomial models to quantify correlation strength [6]



Source: Data Visualization and Business Intelligence with Tab...

46

Interpreting Scatter Plot Patterns

Correlation range: -1 to +1 [7]

Strong correlation: Points follow tight line or curve [7]

Outliers: Points far from main cluster [7]

Identifying Trends

- Points following upward diagonal indicate positive correlation where high values of one variable associate with high values of another [7]
- Downward diagonal patterns show negative correlation, suggesting inverse relationship where one variable's increase corresponds to other's decrease [7]
- Random scatter with no discernible pattern indicates null correlation, meaning variables change independently with no predictable relationship [7]

Finding Outliers

- Look for points significantly distant from main cluster as these represent exceptional cases requiring investigation [6]
- Add reference lines at average values to divide view into quadrants, highlighting high-performing and underperforming segments visually [7]
- Use Detail shelf to add granular dimension like Product ID, creating more data points for thorough correlation analysis without visual clutter [6]

47

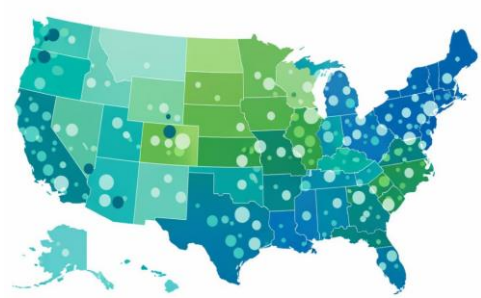
Geographic Mapping

350+ native connectors include geographic data [1]

Auto-geocoding recognizes locations [8]

3 spatial data types: Points, Lines, Polygons [8]

- Tableau automatically geocodes standard geographic entities including countries, states, cities, postal codes, and airport codes without requiring coordinates [8]
- Double-click any geographic field in Data pane to instantly create map with latitude and longitude from built-in database [9]
- Assign geographic roles by right-clicking field name, selecting Geographic Role, and choosing appropriate type [8]
- Connect to spatial files like shapefiles, GeoJSON, or KML for custom geographic boundaries beyond standard geocoding [8]



48

Creating Effective Maps

Point maps
Individual locations plotted [9]

Filled maps
Colored regions showing metrics [9]

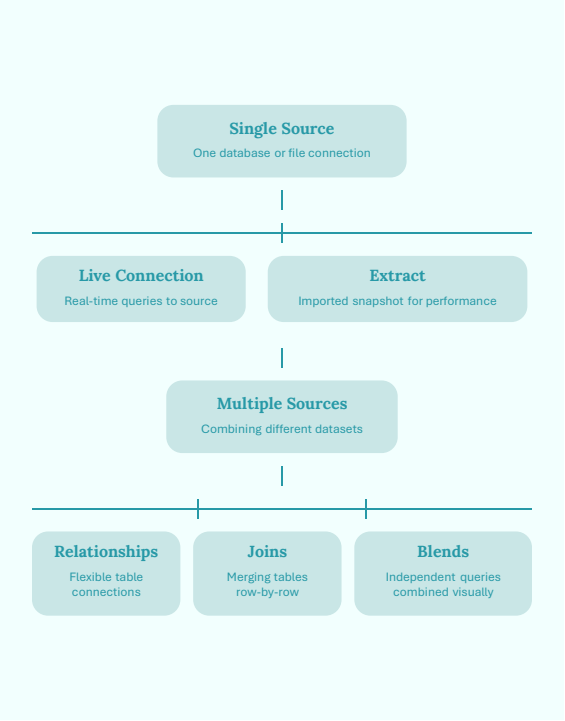
Density maps
Heat patterns for concentration [9]

- For point maps, add measure to Size on Marks card to show proportional circles where larger circles represent higher values geographically distributed [9]
- Change mark type to Map on Marks card to create filled polygons, then drag measure to Color for choropleth maps showing regional differences [9]
- Customize background layers through Maps menu to control visibility of country names, streets, terrain, and base map style matching your design [9]
- Use dual-axis maps to layer different mark types, such as combining filled state polygons with point markers for city-level detail simultaneously [8]

49

Working With Data Sources

- Single data source means connecting to one database, spreadsheet, or file where all tables exist within same connection context [10]
- Live connections query data source in real-time with every interaction, ideal for dashboards requiring up-to-the-minute data accuracy [1]
- Extracts import data snapshot into Tableau's Hyper engine, dramatically improving performance for large datasets and enabling offline analysis [1]
- Choose live when source data changes frequently and performance is acceptable, or extract when working with large datasets or slow databases [1]



50

Method	Timing	Level of Detail	Use Case
Relationships	Before query	Preserved	Different granularity tables
Joins	Before query	Single table	Same granularity merge
Blends	After query	Aggregated first	Published sources

Source: Tableau Relationships vs Joins (& Blends): Complete G...

Combining Multiple Sources

- Relationships preserve table structure, allowing flexible joins that adapt to each sheet's analysis needs [11]
- Joins merge tables by adding columns, but may duplicate data when tables have different detail levels [11]
- Blending queries sources independently, aggregates separately, then combines results without creating unified dataset [10]
- Use relationships for most cases, joins for unified tables, and blends for published sources or different aggregations [11]

51

Applying Table Calculations

FIRST()
Row offset from partition start [14]

LAST()
Row offset to partition end [14]

INDEX()
Current row position number [14]

Running Totals:

Apply Running Total quick calculation to accumulate values progressively, showing cumulative growth over time or across categories sequentially [14]

Percent of Total:

Calculate each mark's contribution to whole using Percent of Total, essential for understanding proportional impact and composition analysis [14]

Ranking:

Use Rank table calculation to identify top performers, with options for dense rank, unique rank, or competition rank depending on tie handling [14]

Year-Over-Year Growth:

Create custom table calculation using LOOKUP function to compare current period with previous year, revealing growth trends and seasonality patterns [15]

Moving Averages:

Smooth fluctuating data by calculating average of specified window, highlighting underlying trends while minimizing short-term variations [14]

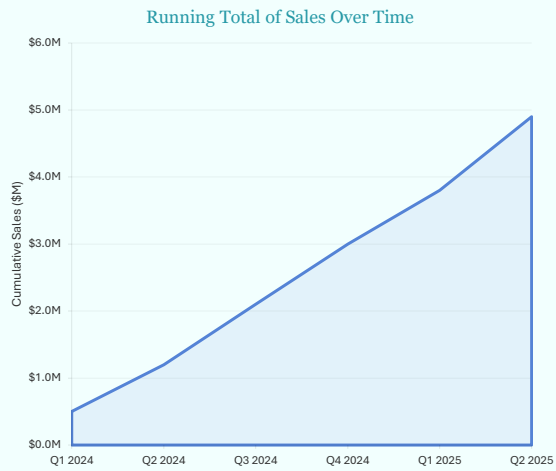
Advanced Functions:

Use WINDOW_SUM, WINDOW_AVG, INDEX, FIRST, and LAST functions to build custom calculations addressing specific analytical requirements precisely [14]

52

Advanced Table Calculations

- Table calculations compute on local aggregated data in Tableau's cache after query executes, operating on results already in visualization [15]
- Quick table calculations accessed via right-click provide instant running totals, percent of total, rank, and moving averages without writing formulas [14]
- Compute Using options control calculation direction: Table Across processes left to right, Table Down top to bottom, and Pane processes within sections [14]
- Advanced settings let you specify addressing (direction) and partitioning (scope) dimensions, determining where calculations restart and how they traverse marks [15]



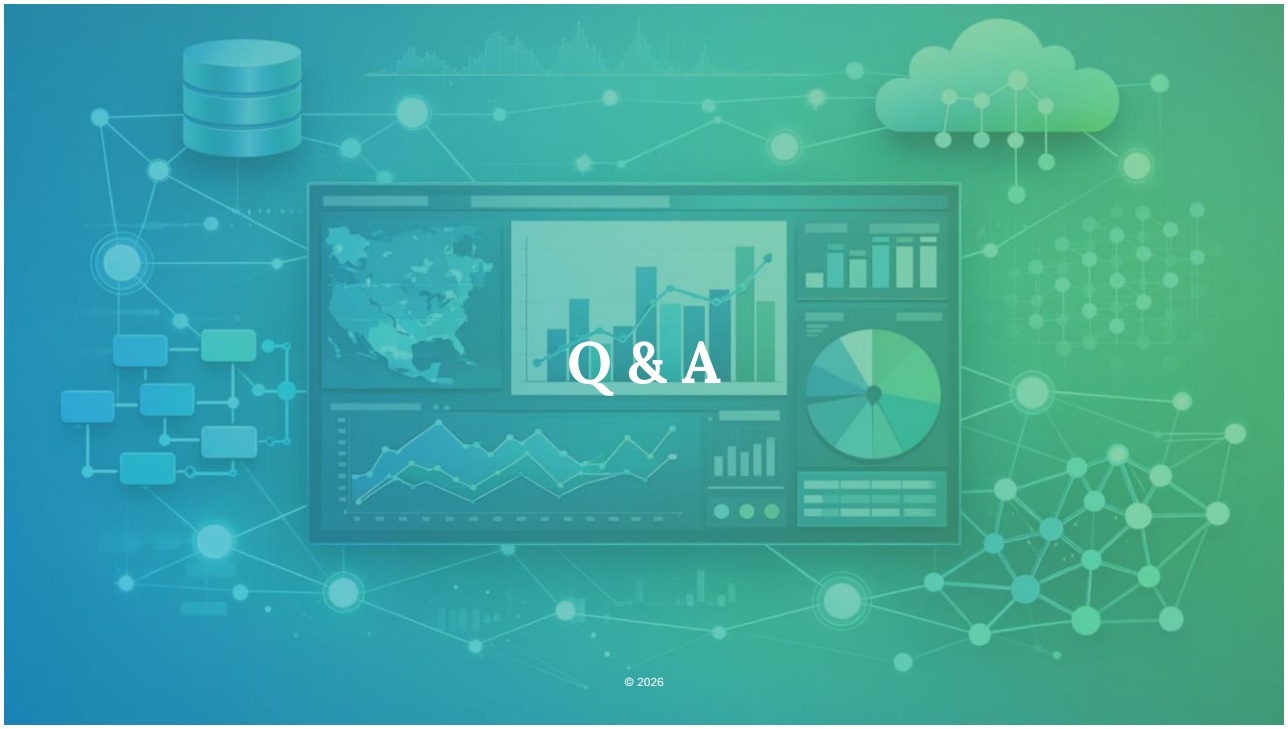
Source: Transform Values with Table Calculations - Tableau

53

References

[1] Introduction to Tableau Desktop
[2] Tableau Desktop Fundamentals Your First Steps in Data...
[3] Data Visualization and Business Intelligence with Tab...
[4] Add Axes for Multiple Measures in Views - Tableau
[5] How to Add Multiple Measure Values in Tableau
[6] Build a Scatter Plot - Tableau
[7] How to Show Correlation in Tableau - thebricks.com
[8] Get Started Mapping with Tableau
[9] Build a Simple Map - Tableau
[10] Blend Your Data - Tableau
[11] Tableau Relationships vs Joins (& Blends): Complete G...
[12] Level of Detail Expressions - Tableau
[13] LOD Expressions in Tableau: A Tutorial With Examples
[14] Transform Values with Table Calculations - Tableau
[15] Learn Advanced Table Calculations using Tableau - Medium

54



55